INCH-POUND

MIL-DTL-2692K
12 July 2024
SUPERSEDING
MIL-DTL-2692J
26 March 2021

DETAIL SPECIFICATION

FLAGS AND PENNANTS

This specification is approved for use by all
Department and agencies of the Department of Defense

1 SCOPE

1.1 Scope. This specification covers the requirements for flags and pennants (other than U.S. National Flags) used by the Department of Defense. The detailed requirements for a given flag will be covered by the drawing for the specific flag.

1.2: Classification. Flags and pennants will be of the following materials and styles as specified (See 6.)

MATERIALS.
- Cotton Bunting
- Nylon, Lightweight Bunting
- Nylon, Heavyweight Bunting
- Nylon-Wool Bunting
- Acrylic Bunting
- Rayon Banner Cloth
- Nylon Cloth, Fluorescent

STYLES (DESIGN APPLICATION)
- Appliqued
- Hand Embroidered
- Machine Embroidered
- Screen Printed (not applicable to acrylic flags)
- Combination of above, as specified

1.2.1 Headings. The heading for the flags and pennants will be of the following types and classes as specified (See 6.2 and 6.5).

Comments, suggestions, or questions on this document should be addressed to The Institute of Heraldry, ATTN: Technical and Production Division, 9325 Gunston Road, Room S-108, Fort Belvoir, VA 22060-5579 or email to usarmy.belvoir.hqda.mil.tioh-webmaster@mail.mil. You may verify the currency of this specification using the ASSIST Online database at https://assist.dla.mil.

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TYPE I - Duck
Class 1 - With distance line (except for acrylic flags)
Class 1A - With distance line (for acrylic flags)
Class 2 - With grommets
Class 3 - With distance line and thimbles
Class 4 – With grommets and leather tabs

TYPE II - Tube and tunnel type (with cotton sheeting lining and leather tab)

TYPE III - Hemmed
Class 1 - With tapes
Class 2 - With leather tabs

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in section 3, and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specification, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents cited in the solicitation or contract.

FEDERAL SPECIFICATION
FED-STD-191 - Textile, Test Methods

FEDERAL STANDARD
CCC-C-430 - Cloth, Sheeting, Cotton, and Polyester and C

COMMERCIAL ITEM DESCRIPTIONS
A-A-59826 - Thread, Nylon
A-A-50196 - Thread, Silk
A-A-1249 - Paper Wrapping, Tissue
A-A-59591 - Cloth, Bunting, Cotton, Mercerized
A-A-59709 - Cloth, Bunting, Nylon and Nylon and Wool
A-A-52094 - Thread, Cotton
A-A-59963 - Thread, Polyester

DEPARTMENT OF DEFENSE SPECIFICATIONS
MIL-DTL-32092 - Leather, Cattlehide, Deerskin and Horsehide, Chrome Tanned
MIL-DTL-32075 - Label, For Clothing, Equipage (General Use)
MIL-DTL-606 - Cloth, Banner, Rayon
MIL-DTL-32629 - Cloth, Banner, Nylon
MIL-DTL-713 - Twine, Fibrous, Impregnated, Lacing and Typing
MIL-F-2310 - Fringe, Ornamental, Rayon
MIL-DTL-3636 - Cords and Tassels, Flag

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MIL-DTL-5038 - Tape, Textile and Webbing, Textile, Reinforcing, Nylon
MIL-DTL-15011 - Pallets, Material Handling, Wood Post Construction, 4-Way Entry
MIL-DTL-15021 - Hooks, Snap Bolt, Swivel-Eye and Rings
MIL-DTL-43566 - Tape, Textile, Cotton or Polyester, General Purpose, Natural or In color
MIL-PRF-17343 - Rope, Nylon
MIL-DTL-22775 - Cloth, Bunting, Acrylic
MIL-DTL-40051 - Leather, Cattlehide, for Glove Leather
MIL-W-44049 - Webbing, Textile and Tape, Textile, Polyropylene, General Purpose, Natural or In Colors
MIL-DTL-226 – Halyards, Signal, Braided Treated

(Copies of these documents are available online at https://quicksearch.dla.mil).

MILITARY STANDARDS
MIL-STD-1916 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-2073-1 - DOD Standard Practice for Military Packaging.

(Copies of these documents are available online at https://quicksearch.dla.mil)).

2.1.2 Other Government document drawings and Publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS
THE INSTITUTE OF HERALDRY (TIOH)
5-1-45A Flag Heading Type I Class1 W Distance line (Except for Acrylic Flags)
5-1-45B Flag Heading Type I Class 2 With Grommets
5-1-45D Flag Heading Type I Class 4 With Grommets and Leather Tabs
5-1-45E Flag Headings Type I Tube and Tunnel Type (with Cotton Sheeting Lining and Leather Tab)
5-1-45F Flag Headings Type III Class 1 and Class 2 Type III Class 1 With Tapes
5-1-45G Flag Heading Type I Class 3 With Distance Lines and Thimbles

THE INSTITUTE OF HERALDRY DOD STANDARD METAL FINISH CHIP SET

(Copies of standards, drawings and Chip sets are available from The Institute of Heraldry, ATTN: Technical and Production Division, 9325 Gunston Road, Room S-108, Fort Belvoir, VA 22060-5579 or usarmy.belvoir.hqda.mi.tioh-webmaster@mail.mil.)

(Figures 1, 2, 3, 4, 5, 6, 7, 8 and 9 are miniature reproductions of the referenced drawings and are attached for information only.)

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2.2 Non-Government publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

THE COLOR ASSOCIATION OF THE UNITED STATES
The Standard Color Reference of America, Tenth Edition

United States Army Color Card, The official Colors for Branches, Agencies, etc.

DOD Standard Color or Official Standardized Shades for Sewing Threads, 1968.

AMERICAN AND EFIRD, LLC
DoD Standard Shades for Heraldic Yarns, 2012 Edition

(Copies for these document are available online at www.salesRA@amefird.com)

(Application for copies of the color cards should be addressed to the Association of the United States, 343 Lexington Avenue, New York, NY 10016-1187.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D5118/D5118M - Boxes, Fiberboard Shipping, Fabrication of
ASTM-D6193 - Stitches and Seams
ASTM-D3951 - Packaging Commercial
NASM16491 - Grommet, Metallic

(Copies for these documents are available online at https://www.astm.org).)

(Copies NASM16491 from the Aerospace Industries Association of America 1250 Eye Street, N.W., Suite 1200 Washington, DC, United States, 20005-3924.https://www.aia-aerospace.org)

2.3 Order of precedence. Unless otherwise noted herein or in the contract. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 Standard Sample. Unless otherwise specified (see 6.2), a standard sample will be furnished by the contracting officer to be used for guidance. Information and as a standard for design and workmanship. Variation from this specification may appear in the guide sample in which event the specification will govern.

3.2 First Article. Unless otherwise specified in the contractor purchase order, a sample will be subjected to the first article inspection (see 4.4 and 6.7).

3.3 Materials. Materials will be as specified herein.

3.3.1 Basic flag materials

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3.3.1.1 Cotton bunting. Cotton bunting will conform to A-A-59591.

3.3.1.2 Lightweight nylon bunting. Lightweight nylon bunting will be conform to Type I, Class A of A-A-59709, At the option of the manufacturer, heavyweight nylon bunting conforming to material cited in 3.3.1.3 may be substituted for the lightweight nylon bunting except when the design is screen printed.

3.3.1.3 Heavyweight nylon bunting. Heavyweight nylon bunting will conform to Type I Class B of A-A-59709 except that paragraph 3.6.1will be deleted: and the following substituted: The finish of the nylon bunting will be equal to or better than the standard sample.

3.3.1.4 Lightweight nylon bunting. Lightweight nylon bunting will conform to MIL-DTL-32629. The finish of the nylon bunting will be equal to or better than the standard sample.

3.3.1.5 Nylon-wool bunting. Nylon-wool bunting will conform to Type II of A-A-59709. As an alternative, a 100 percent polyester fabric of plain weave may be used. The polyester fabric will weight at least 4.5 ounce per square yard with a minimum of 50 yarns per inch in the warp and 32 yarns per inch in the filling.

3.3.1.6 Rayon. The rayon banner cloth will be 50 denier conforming to MIL-C-606. As an alternative, a rayon fabric with the warp using two ply 60 denier (plus or minus 5) and single or two ply 50 denier filling may be used. The number of requirements specified in MIL-C-606 will apply.

3.3.1.7 Fluorescent nylon cloth. Fluorescent nylon cloth will be 200 denier nylon in both warp and filling, 3.0 + 1/2 ounce per square yard, plain weave and heat treated. Fabric will be fluorescent under ultraviolet light.

3.3.1.8 Acrylic bunting. Acrylic bunting will conform to MIL-C-22775.

3.3.2 Duck. The duck used for the heading will be white or in the gray, will weigh not less than 8.25 ounces per square yard, have 2 ply yarn in both warp and filling and will have a breaking strength of not less than 105 pounds in the warp and 100 pounds in the filling direction when tested.

3.3.3 Sheeting.

3.3.3.1 Tube heading lining. Unless otherwise specified, the cotton sheeting used for lining the tube heading will conform to Type I or II, Class 1, Style A or B of CCC-C-430.

3.3.3.2.1 Cotton and mixed nylon-wool flags. Unless otherwise specified, insignia, stars or other symbols for cotton or mixed nylon-wool flags will be made from cotton sheeting conforming to Type II, Class1, Style A of CCC-C 430. The insignia, stars or other symbols may be sized or stiffened to aid in sewing provided the agent does not have a deleterious effect on appearance or serviceability.

3.3.3.2.2 Nylon flags. Unless otherwise specified, insignia, stars or other symbols for nylon flags will be made from nylon conforming to Type I classes A or B of A-A-59709 or nylon conforming to MIL-DTL-32629. The insignia, stars or other symbols may be sized or stiffened to aid in sewing provided the agent does not have a deleterious effect on appearance, color or serviceability.

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3.3.3.2.3 Rayon flags and acrylic flags and pennant. Unless otherwise specified, insignia, stars and other symbols for rayon flags and acrylic flags and pennants will be made from the same material as the flag.

3.3.4 Thread.

3.3.4.1 Cotton. Cotton thread will conform to A-A-52094. The type and sizes of thread will be as specified in Table I.

TABLE I. COTTON THREAD SIZES

Flag or flag component	Type	Ticket No.	Ply	Finish
Cotton bunting flags	I	50	3	Soft
Rayon or fluorescent nylon flag	III	B	3	_
Distance lines	I or II	16	4	Soft or glazed
Cotton duck heading	I or II	10	3	Soft or glazed
Applique of cotton design components nylon-wool flag	I	50	3	Soft

3.3.4.2 Nylon. Nylon thread will conform to A-A-59826. The thread for sewing nylon-wool flags and the hems and reinforcement patches of the nylon flags will be machine twist or twisted bonded multicord, conforming to Type I or II, Class 1, Size B, 2 or 3 ply nylon thread. The thread used for sewing nylon flags, except for the hem and reinforcement patches, will be Type I or II, Size A or B, 3 ply nylon thread.

3.3.4.3 Silk.

3.3.4.3.1. Hand embroidery. The silk thread for hand embroidery willconform to Type IV of A-A-50196 with the following exceptions to the requirements:

Commercial Designation	Minimum Average Length per Ounce (yard)	Minimum Average Break Strength (pound)
A	1000	3.0
FF	233	11.

As an alternative, a 80 mercerized and gassed cotton yarn may be used in lieu of the silk thread and may be substituted as follows:

Silk Thread	Cotton Size
A Type	20/2
FF Type	10/2

3.3.4.3.2 Colorfastness for embroidery cotton yarn. Cotton yarn will show good colorfastness to wet-dry cleaning, water, light and crocking.

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3.3.4.4 Rayon. Rayon thread used in Bonnaz machine embroidery will be 600 denier, continuous filament, and regenerated cellulose (viscose or cuprammonium) rayon thread. The thread will have not less than 460 yards per ounce and a breaking strength of not less than 2.3 pounds.

3.3.4.5 Polyester. Polyester thread for sewing acrylic flags and pennants will conform to Type I, Class 1, Subclass A, Tex Size 45 or Letter size B, 3 ply of A-A-59963 and will be white in color.

3.3.4.6 Color and colorfastness. Unless otherwise specified, the color of the thread will match the color of the material on which it is used. Except for reinforcement patches, multicolored flags and pennants made of cloth other than acrylic bunting may be sewn with white thread or thread of the predominant color. Acrylic flags and pennants will be sewn with thread. The thread for duck heading will be white or unbleached. When a fringe is specified, the color of the attachment threads for multicolored flags, visible on the reverse side of the flag will match the predominant color of the base material. The colorfastness requirements of the thread used will be equal to or better than the material on which it is used as described in 3.4.1.1.1.

3.3.5 Distance lines. The distance lines will be 1/4 inch (0.64cm) in diameter nylon rope conforming to MIL-PRF-17343 or 1/4 inch (0.64cm) in diameter twisted or solid braided polypropylene rope having a minimum breaking strength of 1000 pounds as shown on the applicable drawing.

3.3.5.1 Twine. The twine used in wrapping the free end of the distance line will be waxed twine conforming to Type N, Class 3 of MIL-DTL-713.

3.3.6 Tape. Tape will be 1/2 inch (1.27cm) nylon tape conforming to Type III of MIL-DTL-5038.

3.3.7 Snap hook and ring. The snap hook and rings will conform to MIL-DTL-15021.

3.3.8 Grommets. Grommets will conform to Type II, Trade Size 2 of NASM16491 or Type III, Trade size No. 2 of NASM16491. Grommets will have a bright finish.

3.3.9 Leather. Leather for the tabs conform to Type I, Class2 of MIL-DTL-32092, 3 to 4 ounces when measured on a Woburn Gauge or Type II of MIL-L-40051. When natural colored leather is used, the crocking requirements will not apply.

3.3.10 Fringe. Except for automobile flags and unless otherwise specified, the fringe will conform to Class 2 of MIL-F-2310. The fringe for automobile flags, when specified, will conform to Class 1 of MIL-F-2310. The color of the fringe will be as specified on the applicable drawing.

3.3.11 Webbing. When specified, the webbing used for reinforcement of Type I, Class 3 heading will conform to Type I, Class 1 or 2 of MIL-DTL-43566 and will be of the width specified on the applicable drawing.

3.3.12 Cord and tassels. When specified (see 6.2), cord and tassels will conform to MIL-C-3636 and applicable drawing. Unless otherwise specified on the drawing

3.3.13 Thimbles. Thimbles will be galvanized steel and of the type used for 1/4 inch (0.64cm) in diameter wire rope. They will be semicircular at the closed end and tapering at free ends.

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3.4 Design and tolerance. The design and size of the flags and pennants will be as specified on the applicable drawing, a tolerance of plus or minus 5 percent will be allowed for all dimensions exceeding 2 inches (2.54cm) except that maximum and minimum measurement of dimensions on a single flag may not differ more than 2 percent of the required dimensions and both be within 5 percent tolerance. For dimensions 2 inches (2.54cm) or under, a tolerance of plus or minus 1/8 inch (0.32cm) will be allowed.

3.4.1 Components of design. Components of the design will be applied by one or a combination of the methods listed below and as specified on the applicable drawing. The face side of the flag will be designated as the side visible when the heading is on the viewers left Unless otherwise specified on the drawing, letters and numerals will read from left to right on both sides of the flag.

(a) Applique. Appliqued design will be identically oriented in the location shown on the applicable drawing and will appear proper on each side of the flag. Unless otherwise specified and except for acrylic flags or where components of design are printed on percale, the thread used for appliqued with thread matching the basic color of the percale. Components of design for acrylic flags and pennants will be appliqued with white polyester thread specified in 3.3.4.5.

(b) Hand Embroidery. Hand embroidered components of the design will be embroidered in the location shown on the applicable drawing using thread specified in 3.3.4.3. Components of the design will appear proper on the face side of the flag, be clearly distinguishable and show identical on the opposite side of the flag as if printed through the material.

(c) Machinery embroidery. Machine embroidered components of the design will be embroidered in the location shown on the applicable drawing using thread specified in 3.3.4.4. Components of the design will appear proper on the face side of the flag, be clearly distinguishable, symmetrical through the material.

(d) Screen printing. Screen printed components of the design will be printed in the location shown on the applicable drawing using screens made by the photographic method. Components of the design will appear proper on the face side of the flag, be clearly distinguishable, symmetrical and show identical on the opposite side of the flag as if printed through the material. Acrylic flags will not be screen printed.

3.4.1.1 Color. Colors will be as specified on applicable drawings and will conform to approved shade standards cited in the latest edition of the standard Color Reference of America (see 2.2). If cable numbers cited on applicable drawings do not match those cited in the Standard Color Reference of America, the outdated number will be converted to the current cable number. When cable number 70001 is cited, the number will be converted to 80001. The colorant system for screen printed design will maintain the essential textile appearance of the base material. The dye will not adversely affect the hand of the fabric. Coating will not be used unless approved by the contracting officer. When prints are required to register, the design will be aligned when viewed by through lighting. When prints are applied to the fabric for subsequent applique, some strike-through of the print is permissible provided it does not adversely affect the appearance of the finished flag or pennant. When penetrating dye is used, the color on the face will match the color specified on the applicable drawing. A slight change of the color will be allowed on the reverse side.

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3.4.1.1.1 Colorfastness. The colorfastness of the base material for the flags and pennants will conform to the requirements of the applicable specification for the material. Colorants for screen printed designs will provide colorfastness equal to or better than the dye base fabric. The colorfastness to crocking for napthol colors or black areas may be fair. Rating on an adjective basis will be performed only when a standard sample of the exact material is not provided.

3.5 Construction. Flags and pennants will be constructed in accordance with the applicable drawing and the requirements specified herein.

3.5.1 Basic Construction. All raw edges will be hemmed. A selvage end may be used at the top or bottom edge except that the flags and pennants manufactured from nylon, acrylic bunting or rayon banner cloth will be hemmed on all edges. Hems will be turned to the back side of the flags except hems or fringed flags will be turned to the front side of the flag. The ends of all stitching will be backstitched not less than one inch (2.54cm) when not caught in a seam or other stitching. Except for flags and pennants of acrylic bunting, when the size of the pennant necessitates the piecing of the basic material, the seam, whenever possible, will be in the center two – thirds of the material, the base cloth may be used with the wrap running in the vertical or hoist except the pennant code flags representing the letter Y, M and V and the numerals 4, 5, and 6 will have material for the strip cut on a bias and set in on the straight of the material.

3.5.2 Headings. Unless otherwise shown on the applicable drawing, headings for flags and pennants will be as specified herein.

3.5.2.1 Type I (Duck). Type I heading will be fabricated from duck specified in 3.3.2 and the requirements specified herein.

3.5.2.1.1 Class I (Heading with distance line (except for acrylic flags). Type I, Class I headings will be constructed in accordance with the details and dimensions shown on TIOH drawing 5-1-45A (Figure I) and will meet the following requirements:

(a) The heading will be made by cutting a piece of duck 6 inches (15.24cm) wide and 2 inches (5.08cm) longer than the hoist of the flag. The edges and ends of the duck strip will be turned-in 1 inch (2.54cm) and the duck doubled over to form a 2 inch (5.08cm) wide heading.

(b) The hoist end of the flag will be turned-under 3/4 inch (1.91cm) and may be sewn with a single row of stitching. The folded end of the flag will be then be inserted between the edges of the duck for a distance of 1 inch (2.54cm) so the ends of the flag will not exposed. With the turned-in sides and ends of the duck even, the hoist end of the flag and the open side of the heading will be stitched with a minimum of three rows of equidistant stitching 1/4 (0.64cm) to 3/8 inch (0.95cm) gauge (1/8 inch (032cm) stitching margin).

(c) The flag or pennant will have a distance line specified in 3.3.5 running through the heading. The line at the top of the heading will pass through a ring specified I 3.3.7, double back and superimposed upon itself and sewn with zigzag stitches through both thicknesses of the line for a minimum distance of 2 inches (5.08cm). The line will then drawn into the heading until the ring is approximately flush with the outer edge of the heading. The top of the heading will then be closed by two rows of stitches 1/4 (0.64cm) to 1/2 (1.27cm) gauge, either up to or through the halyard. The distance line will be secured within the heading by zigzag stitches at the top, center and bottom for minimum of 2 inches (5.08cm) at each location. The lower end of the heading will then be closed by two rows of stitching in the same manner as the top.

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(d) The snap hook specified in 3.3.7 will be attached to the distance line by passing the line through the swivel loop, double back and superimposed upon itself for a distance of not less than 2 inches (5.08cm) and sewn by zigzag stitching for a minimum of 2 inches (5.08cm). Zigzag stitching at the top and bottom of the line will pass through both thicknesses of the distance line. Prior to zigzag stitching, the free end of the end of the distance line will be wrapped for a distance of not less than 1/2 inch (1.27cm) with not less than 8 turns using waxed twine specified in 3.3.5.1 The overall length of the extended portion of the distance line snap hook, measured from the lower edge of the heading, will be 9 inches (22.86cm) for signal flags and pennants and 24 inches (61cm) for all others.

(e) The circumference of distance lines for flags and pennants having an area of less than 16 square feet will be 3/4 inch (1.91cm). Unless otherwise specified on the applicable drawing, the circumference of the distance lines for the flags and pennants with the area of 16 square feet and larger will be 1 inch (2.54cm). The stich type will be as specified in table II. At the option of the manufacturer, a type I, class 1A heading may be used as an alternate method of construction.

3.5.2.1.2 Class IA (Heading with distance line for acrylic flags). Type I. Class IA headings will be constructed as specified in 3.5.2.1.1 with the following exceptions.

(a) The heading will be constructed in accordance with the details and dimensions of TIOH drawing 5-1-45H (figure 2). The heading will be stitched to the hoist end with four rows of stitching and a stitching margin of 1/8/inch (0.32cm). The stitching gauge will be 1/4 inch (0.64cm) to 3/8 inch (0.95cm).

(b) The distance line will be fabricated from 1/4 inch (0.64cm) diameter nylon rope specified in 3.3.5.

(c) The width of the zigzag stitching will cover no less than one half the diameter of each strand of rope. The end of the zigzag stitching will be firmly secured by backstitching or overstitching or by increasing the number of stitches per inch at the ends for a minimum of 1/4 inch (0.64cm).

(d) The turned back end of the distance line will be zigzag stitched from the snap hook to the free end of the distance line within the heading.

(e) The free end of the nylon rope will be wrapped as specified in 3.5.2.1.1 or may be taped or fused at the ends to prevent raveling. Both cutting and fusing of the nylon rope may be accomplished in one operation if a hot wire blade is used for the operation.

3.5.2.1.3 Class 2 (Heading with grommets). Type I, Class 2 heading will be constructed in accordance with the details and dimensions shown on TIOH drawing 5-1-45B (figure 3) and will meet the following requirements. Stitch type will be as specified in Table II. The heading will be made with a piece of duck 6 inches wide and 5 inches (12.7cm) longer than the flag hoist to which it is to be attached. Both side edges will be turn-in I inch (2.54cm) and both ends turned-in 2-1/2 inches (6.35cm). The hoist end of the flag will be turned under3/4 inch (1.91cm) and may be sewn with a single row of stitching. The folded ends of the inserted flag hoist will not be exposed. The raw edges of the ends of the duck strip will be closed with a single row stitching using a 1/8 inch (0.32cm) stitching margin or as an alternate, the row of stitching may be placed after the duck is double over. The duck strip will then be doubles over to form a finished heading 2 inches (5.8cm) wide. With the sides and ends of the duck strip even, two row of stitching (not

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less than 3/16 inch (0.48cm) stitching gauge, 1/8 inch (0.32cm) stitching margin) will be sewn through all thicknesses. Single rows of stitching will be placed along the centerfold edge and ends of the duck strip using a 1/8 (0.32cm) to 1/4 inch (0.64cm) stitching margin. The duck heading will be finished by securely clinching two No. 2 grommets
Specified in 3.3.8 in positions shown on figure 3 for type I, Class 2 heading.

3.5.2.1.4 Class 3 (Distance line with thimble). Type 1, Class 3 headings as shown on TIOH drawing 5-1-45G (figure 4), shall be constructed in accordance with the dimensions specified for Class 1 headings in 3.5.2.1.1 and shown on TIOH drawing 5-1-45A (figure 1). The class 3 headings shall be fitted with galvanized steel thimbles specified in 3.3.13 to provide a loop or eye not less than 1-1/2 inches (3.81 cm) in length.

3.5.2.1.5 Class 4 (Heading with grommets and leather tabs). Type I, Class 4 heading will be constructed in accordance with the details and dimensions shown on TIOH drawing 5-1-45D (figure 5) from duck specified in 3.3.2 and will meet the following requirements:

(a) The heading will be made by cutting a piece of duck 8 inches (20.32cm) wide and 5 inches (12.7cm) longer than the hoist of the flag. Stitch type will be as specified in Table II.

(b) The hoist end of the flag will be positioned 1 inch (2.54cm) in from the side edge of the duck, 2-1/2 inches (6.35cm) from each end and stitched with one row of stitching. The side edge of the duck attached to the flag will turned-in 1 inch so that a 1-inch (2.54cm) strip of the flag material the same length as the hoist will be folded inside the turned-in duck fold as shown on the applicable drawing. The opposite side edge of the duck will be turned-in 2-1/2 inches (6.35cm). The raw edges of the top and bottom duck ends will be stitched closed with single row of stitching with a 1/8 inch (0.32cm) stitching margin.

(c) A tab, made of leather specified in 3.3.9, 3/4 inch ± 1/16 inch (1.91cm + 0.16cm) wide and not less than 2 inches (5.08cm) long, having a slit and eyelet 3/4 inch ± 1/16 (1.91cm ± 0.16cm) long, for fastening the tab over a knob or button of the flagstaff will be centered and placed flush with each end of the heading as shown on TIOH drawing 5-1-45D (figure 5). The tabs will be secured through the two thicknesses of duck by box stitching as shown on the applicable drawing. Stitching will be securely backstitched.

(d) Four No. 2 grommets specified in 3.3.8 will be clinched into position as indicated on the applicable drawing. The duck heading will be doubled over to form a heading width of at least 2-3/4 inches (6.99cm). Position the sides and ends of duck evenly, with the grommet holes aligned and sew two rows of stitching not less than 1/16 inch (0.16cm) gauge, 1/8 inch (0.32cm) margin through all thicknesses.

3.5.2.2 Type II (Tube and tunnel type heading with cotton sheeting lining and leather tab. Type II heading will be constructed in accordance with the details and dimensions shown on TIOH drawing 5-1-45E (figure 6) from cotton sheeting specified in 3.3.3.1 and will meet the following requirements.

(a) The heading will be made by cutting a piece of sheeting 6 inches (15.24cm) wide and 5 inches (12.7cm) longer than the hoist of the flag; guidon sheeting will be cut down proportionately in the width so a 2-1/4 inches (5.70cm) heading is formed. Each end will be folded under 2-1/2 inches (6.35cm).

(b) The sheeting will be fitted with a tab made of leather specified in 3.3.9 at each end. Each tab will be 3/4 inch ± 1/16 inch (1.91cm ± 0.16cm) wide and not less than 2 inches (5.08cm) long. Each tab will have a slit and eyelet 3/4 inch ± 1/16 inch (1.91cm ± 0.16cm) long

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for fastening over a knob or a bottom on the flagstaff. The slit and eyelet will be centered on the width of the tab shown on the applicable drawing. The tab will be placed within the heading and centered opposite the stitching of the hem. The tab will be secured through the two thicknesses of lining by box stitching as shown on the applicable drawing. As an alternate, the leather tab may be secured by stitching through both lining and heading material. The stitching will be securely backstitched. The contractor's label specified in 3.8 will be affixed at the same time the tabs are placed on the sheeting.

(c) The sheeting strip will be placed on the back of the flag at the hoist end with the folded ends of the strip flush with hemmed or salvage edges of the flag at the top and attached thereto by a single row of stitching across each folded end of the strip. The raw edge of the stitching will be placed 1/2 inch (1.27cm) from the raw edge of the flag. The tube or tunnel will then be formed by hemming the hoist end of the flag with the sheeting strip inside. Sheeting may also be seamed separately with 1/2 inch (1/27cm) seam to form a tunnel. Turn under hoist edge of flag 1/2 inch (1.27cm) and insert under the folded side edge of sheeting strip. Turn under and stitch with a single or double row of stitching (3/16 inch (0.48cm) margin), which will pass through both the turned edge of the flag and both edges of the unseamed or seamed lining strip. The flat width of finished hem will be not less than 2-3/4 inches (6.99cm) (except for guidons which will not be less than 2-1/4 inches (5.70cm).

3.5.2.3. Type III (Flags without duck or cotton sheeting heading). Type III headings will have the hoist end of the flag hemmed and will be constructed in accordance with the requirements specified herein.

3.5.2.3.1 Class 1 (with tape). Type III, Class 1 heading will be constructed in accordance with the details and dimensions shown on TIOH drawing 5-1-45F (figure 7). Stitch type will be specified in two rows of stitching. Nylon tape specified in 3.3.6 will be cut 12 inches long (3048cm), doubled end-to-end so as to form tie-tapes and attached to the hem and the other 1 inch (2.54cm) from the bottom. The tapes will overlap the full width of the hem and be arranged so as to facilitate the tying of the flag to a flagstaff.

3.5.2.3.2 Class 2 (with leather tabs). Type III, Class 2 heading (automobile and other flags when required) will be constructed in accordance with the details and dimensions shown on TIOH drawing 5-1-45F (figure 7). Two rectangular leather tabs 1-3/8 inch (3.49cm) long and 5/8 inch (1.59cm) wide, will be located and secured within the hem as shown on applicable drawing. After insertion of the tabs, the hoist end will be hemmed.

3.5.3 Stitching of fringe. When fringe is specified for flags, a continuous length of the fringe conforming to 3.3.10 will be stitched to the front side of the flag at the top, fly and bottom edge of the flag and completely around each end of the heading. The fringe will be neatly turned under and caught in the heading stitching. Selvage edges at top and bottom will be hemmed with Type EFa-1 stitching, cut edges with EFb-1 stitching. The stitching used to attach the fringe may be incorporated in the information of the hem. As an alternative, edges may be hemmed with an EFa-1 stitching and turned over prior to attachment of fringe. The fringe will be superimposed on the hem and evenly attached around the flag with a single row of stitching through the center of the fringe using seam type LSa-1, except that automobile flags will have fringe attached with three rows of stitching, seam type LSa-1.

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3.6 Stitches, seams and stitching. Unless otherwise specified, stitches, seams and stitching will conform to ASTM-D6193 and shown in Table II.

TABLE II. STITCHES, SEAMS AND STITCHING

Section of flag or pennant	Type of seam or stitching	Minimum hem width (inch)	Stitch type	Stitches-per inch
Hem (Top and Bottom):				
Flags with Fringe……	EFa-1 or EFb-1	3/32 – 3/16	301	12
Flags, nylon……	EFa-1 or EFb-1	3/16	301	12
All other flags **1/**…….	EFa-1 or EFb-1	3/8	301	12
Tail or fly end:				
Flags with fringe……..	EFb-1	3/32 – 3/16	301	12
Semaphore, wigway and flags with Type II heading...	EFb-2	1/2	301	12
All other flags……..	EFd-3 or EFb-4	3/4	301	12
Reinforcement patches….	LSd-1 or LSd-2	-	301	12
Seams: **5/**				
Cross and fly……..	LSc-2	-	301	12
Bias…….	LSr-2 or LSc-2	-	301	12
Appliqued design:				
Cotton **3/**…….	-	-	304	24
Nylon, Nylon-Wool, Acrylic **4/**……..	-	-	304	28 to 32
Rayon **2/**.....	-	-	304	58
Headings:				
Type I (Ducks);		(Heading Width)		
Class 1……..	3.5.2.1.1	2	301	6
Class1A……	3.5.2.1.2	2	301	6
Class 2……..	3.5.2.1.3	2	301	6
Class 3……..	3.5.2.1.4	2	301	6
Class 4……..	3.5.2.1.5	2-3/4	301	6
Type II (Tunnel type) for all flags except guidons……..	3.5.2.2	2-3/4	301	12
Guidons…….	3.5.2.2	2-3/4	301	12
Type III (Hemmed):				
Class 1……	3.5.2.3.1	1/2	301	12
Class 2……	3.5.2.3.2	1-1/4 ± 1/4	304,308 or	
Distance lines…….	-	-	hand stitched	6
Fringe……	LSa-1	-	301	12

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1/ Seam types of the same letter designation (e.g., EFb, LSd) are the same seams and differ only in the number of rows of stitching as designated by the number following the dash (e.g., EFb-1, EFb-2 or LSd-1, LSd-2).
2/ Throw of stitch 1/16 inch (0.16cm) (unless otherwise specified on drawing).
3/ Throw of stitch 1/8 inch (0.32cm) (unless otherwise specified on drawing). A 1/16 inch (0.16cm) throw is permitted if stitching area is restricted in size.
4/ Throw of stitch 1/16 (016cm) to 1/8 inch (0.32cm) (unless otherwise specified on drawing).
5/ As an alternative, seams of signal flags and signal pennants may be stitched with a combination of SSa-2 seam and LSa-2 seam. After completion of the SSa-2 seam with stitch type 515, the edge will be smoothed and the seam type LSa-2, stitch type 301, will be applied within 1/8 inch (0.32cm) from the edge with a 1/4 inch (064cm) gauge. The throw of the SSa-1 seam will be 3/8 inch (095cm). The sSSa-2 and LSa-2 will have a minimum of stitches per inch.

3.1 Reinforcement patches. When required, all patches will be located on the reverse side of the flag or pennant and will be fabricated of the same material in the same color as that of the section to be reinforced. All exposed edges will be turned under a minimum of 1/4 inch (0.64cm) and securely stitched. Reinforcement patches are required on the following

(a) Flags with type II and III heading.

(b) Flags with a hoist of 24 inches (60.96cm) or less.

(c) Pennants made from acrylic bunting.

(d) International signal flags made from acrylic bunting except for size 3, 4 and 6.

3.1.1 Cotton and nylon-wool bunting flags. Reinforcement patches on cotton and nylon-wool bunting flags with Type I headings will be sewn to each corner at the hoist end of the flag and will extend within the heading the same distance as the flag base. The size of the reinforcement patches will be as shown on the applicable drawing or at the option of the manufacturer, the patches may be 4 inches (10.16cm) square for flags with hoist lengths of 2 feet (60.96cm) to 3 feet (91.44cm) or 6 inches (15.24cm) square for flags with hoist lengths over 3 feet (91.44cm). The size of the patch will be as near the required size as possible.

3.1.2 Nylon and acrylic bunting flags. Reinforcement patches on nylon and acrylic bunting flags with Type I headings will be sewn in each corner of the fly end of the flag. The patches will be in the shape of right angle triangles with the apex (90 degree angle) located at each corner of the fly end of the flag. The size of the reinforcement patches will be shown on figure 9. Sewing of the patches to the fly end of the flag will be accomplished by folding the top or bottom edge (as applicable) of the flag base 3/8 inch (0.95cm) (see figure 8) and stitching 1/16 inch (0.16cm) from the inner and outer edges of the fold to form the top and bottom hems. The fly end hem will be formed by folding the edge of the base material 3/4 inch (1.91cm) over the patch and then refolding both plies of fabric over 3/4inch (1.91cm) (see figure 8) and stitching the hem with three evenly spaced rows of stitches. The outer rows of stitching will be not more than 1/16 inch (0.16cm) from the outer edges of the hem.

3.2 Marking. Flags and pennants with Type I headings will be marked on the heading in accordance with Type IV, Class 9 label of MIL-DTL-32075 with letters appropriate to the size of the heading but not more than 3/4 inch (1.91cm) high. The lettering will run along the length of the heading, beginning immediately below the upper grommet on class 2 headings and in corresponding position on all other Type I headings. Marking will contain information required

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for Class 5 labels and will be equal to Class 8 labels in fastness of printing. For Type II heading, a label conforming to Type I, Class 5 of MIL-DTL-32075 will be affixed adjacent to the top leather strap.

3.3 Workmanship. Buttons will be clean, well made and will meet the acceptable quality levels established by this specification. All stitching which is not caught up by a seam or stitching will be backstitched a minimum of one inch (2.54cm). Thread tension will be maintained throughout and all thread ends will be trimmed.

4 VERIFICATION

4.1 Inspection responsibility. Unless otherwise specified in the contract or order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that the products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorized submission of know defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality Conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.5.

4.4 First article inspection. In addition to the quality assurance provisions of the subsidiary specifications and standards, inspection shall be performed on components and materials listed in table V for test characteristic shown. The inspection level shall be S-1. The requirements are applicable to the lot average.

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TABLE III. COMPONET TESTING

Component	Characteristic	Reqm,t Ref¶	Test Method	Reqm,t Individual Unit	Appl to Lot Average	#Deter Per Sample Unit
Silk Embroidery Thread	Breaking Strength length	3.3.4.3.1	4104	-	X	-
	Weight		4010	-	X	-
Cotton yarn	Color Fastness to Wet-dry Cleaning	3.3.4.3.2	5622	-	-	-
	Water		5630	-	-	-
	Light		5660	-	-	-
	Crocking		5661	-	-	-
Bonnaz Rayon embroidery Thread	Ply **2/** Breaking Strength	3.3.4.4	Visual	X	-	1
	Length-Weight		4102	-	X	-
	Material **2/**		4010 Std	-	X	-
	Identification		Comm'l	X	-	1

1/ Test will be in accordance with FED-STD-191.
2/ Results reported as pass or fail.

4.5.2 In-process inspection. Inspection shall be performed at any point or during any phase of the manufacturing process to determine whether operation or assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated non-conformance. In-process inspection shall be conducted to see that accomplishment of the following is in accordance with the specification requirements.

Requirements, operation or assembly	Requirement paragraph
Heading construction	3.5.2 thru 3.5.2.3.2 as applicable
Machine thread shall be checked at The machine to determine that threads, As specified, are used	3.3.4 thru 3.3.4.6 as applicable
Turn-in and turn-under of seams,	3.5.2 thru 3.5.2.3.2

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Hems, reinforcement patches and Headings.	and 3.7 thru 3.7.2, as applicable

4.5.3 Intermediate testing. Testing of the partially fabricated flag or pennant shall be performed in accordance with Table IV for characteristic shown therein. The sample unit shall be one screen printed design for appliqued flags or one screen printed flag base as applicable, prior to sewing into the completed flag. The inspection level shall be S-1 and the acceptable quality level (AQL) for each characteristic shall be 6.5 defects per hundred units. All requirements shall be applicable to the sample unit.

TABLE IV TESTING OF THE INTERMEDIATE PRODUCT.

Screen printed design for	Colorfastness to:	Requirement paragraph	Test method 3/
Cotton, nylon and nylon-wool flags	Light Crocking Water	3.4.1.1.1 3.4.1.1.1 3.4.1.1.1	5660 1/ 5651 5630 2/
Rayon	Light Crocking Water Wet-dry Cleaning	3.4.1.1.1 3.4.1.1.1 3.4.1.1.1 3.4.1.1.1	5660 1/ 5651 5630 2/ 5622

1/ In the event of a dispute resulting from test method 5660, or as a result of suspected anomalous behavior of certain dye types or formulations, the contracting officer may authorize the exposure to natural light in accordance with method 5662.
2/ Method 5630 shall be used except the test temperature shall be 23.9 degrees to 29.4 degrees C (75 degrees to 85 degrees F).
3/ Test method in accordance with FED-STD-191.

4.5.4 Inspection of the end item.

4.5.4.1 Visual examination of flags and pennants. Examination shall be made at a distance of approximately 18inches (45.72cm) to 40 inches (101.6cm) with good lamination. The defects found during examination shall be classified in accordance with 4.5.4.1.1 and 4.5.41.2. The lots shall be inspected in accordance with 4.5.4.1.3. The unit of product for these examinations shall be one completely fabricated flag or pennant as applicable.

4.5.4.1.1 Examination of flags or pennants for defects in material, design, construction, workmanship and marking. Defects designated by an asterisk (*) shall be classified as a major when seriously affecting serviceability or appearance and minor when affecting serviceability or appearance, but not seriously.

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TABLE V. DEFECTS

EXAMINE	DEFECT	CLASSIFICATION		
		MAJOR	(*)	MINOR
Quality of:				
Type	Any component not fabricated of specified material……..	-	*	-
Color	Basic color other than specified……..	X	-	-
	Off shade……	-	*	-
Material	Hole, cut, tear, smash or weak place…….	-	*	-
	Spots, stains, streaks or shade bars (considering the use position)…….....	-	*	-
	Two or more continuous broken or missing ends or picks in cotton, acrylic, nylon and nylon-wool cloth……	X	-	-
	Two or more continuous broken, missing ends or picks in rayon cloth……	X	-	-
	Double pick, coarse pick or mispick clearly noticeable……	-	- *	-
	Slub, slough-off, float or skips……	-	*	-
Quality of Distance lines:				
Types				
Braiding	Not fabricated of specified material……...	-	-	X
	Uneven or loose resulting in weak place……	X	-	-
	Any cut, chafing, fraying, or abrasion……	-	*	-
Quality of metal components:				
Snap hook & ring				
	Not fabricated of specified material……	-	-	X
	Snap hook does not work freely; binds or catches when being opened or closed……	-	*	-
	Broken, Cracked or malformed……	-	*	-
Grommets				
	Broken, cracked or malformed……	-	*	-
	Contains rough or sharp edges……..	-	-	X
Quality of leather components	Not fabricated of specified leather……..	-	-	X
	Hole (other than required), cut, tear, or weak place.			
	Broken, cracked or malformed……	-	*	-
Quality of fringe, cord and tassels	Material, type, construction, color, or color combination, not specified……			
	Frayed edges…….			
	Spotted, Stained or soiled…….	-	*	-
	Cut or tear…….	-	*	-

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	TABLE V DEFECTS CONTINUED			
Screen Printing				
	Wood mold not completely covered with braiding..	-	-	X
	Wood mold loosely or unevenly covered……...	-	*	-
	Any part of design omitted…….	X	-	-
	Coating used in lieu of die……..	-	*	-
	Poor registration, over printing or feathering…….	-	-	-
	Wrong color, or arrangement of colors……..			
	Dye used in screen printing adversely affects hand of fabric……..	-	*	-
Construction and workmanship (General)	Dye bronzed or mottled to extent of affecting appearance at 6 feet (182.88cm)……	-	*	-
	Poor penetration of dye from front to back……	X	-	-
	Operation omitted or not accomplished as specified..	-	*	-
	Flag pieced where piecing is prohibited…….	-	*	-
	Component is not the required type, class or size……	-	*	-
Stitches, seams and stitching	Warp threads of the basic material of the flag in other than the fly direction of the flag, except as permitted in 3.5.1……...	-	*	-
	Any component missing……	-	*	-
	Wrong type of seam, stitches or stitching……..			
	Wrong color of thread……	-	-	X
	Row of stitches partially or completely missing……	-	*	-
	Wrong stitching gauge or margin……..	X	*	-
	Seams improperly formed, material caught in unrelated stitching operation……..		*	-
	Ends of stitching not backstitched one inch (2.54cm) (when not caught in other seams or stitching……..	X	-	-
	Run-offs resulting in open seam……	-	-	X
	Needle chews……..			
	Any raw edges exceeding 1/2 inch (1.27cm) in length……..	X	-	-
	Loose stitch tension resulting in a loosely secured seam or thread lying on surface of seam……	X	-	-
	Tight stitch tension (stitches break when normal pull is exerted in lengthwise direction of seam or stitching……..	X	-	-
	Stitches per inch for seam, hems and reinforcement patches-four or more stitches less than specified…..	-	-	X
	--one to three stitches less than specified……..	X	-	
	Stitches per inch for duck heading:			X
	--two or more stitches less than specified……..	X	-	
	--one stitch less than specified……	-		-
	Stiches per inch for appliqueing:			
	--five or more stitches less than specified……..			

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	TABLE V DEFECTS CONTINUED			
	--one to four stitches less than specified……	-	-	X
	NOTE: Variation in the number of stitches per inch caused by the operator speeding up the machine or pulling fabric, such as sewing over heavy places, or in turning corners shall be classified as follows:			
	Major to minors, minors to no defect. Seam twisted, pleated or puckered……	-		
	More than two continuous skipped stitches……...	X	-	-
	Broken stitches in heading seam causing an unstitched length of one inch or more……	-	*	-
	Broken stitches in heading seam causing an unstitched length of less than 1/2 inch (1.27cm)……	-	–	X
Embroidery	Broken stitches……...	-	*	-
Design (insignia,	Not applied as specified…….	X	-	-
emblem, stars,	Design or parts of design reversed or inverted……..	X	-	-
lettering, etc.)	Design misplaced or in wrong location…….	-	*	-
	Lettering does not read from right to left on both side (unless otherwise specified)……	X	-	-
	Design not oriented on each side as specified……	X	-	-
	Pattern marks used for location of design visible on finished product……...	-	*	-
	Design or insignia not neatly trimmed……...	-	*	-
	Evidence that sizing or stiffening agent used as an aid in sewing has deleterious effect……	-	*	-
Hems	Nylon or acrylic flags not hemmed on all edges……	X	-	-
	Hem uneven……	-	*	
	Hem on wrong side of flag…….	-	-	X
	Hem puckering……...	-	*	-
	Positioned on face side of flag……...	-	-	X
Reinforcement patches	Portion of patch at heading not inserted within the heading not inserted within the heading as specified.			
	Patch not the same material or color as section reinforced……	-	*	-
	Size or shape of patch not as specified……..	-	*	-
	Not completely attached as specified…….	-	-	X
Fringe	Ends of fringe not caught in heading stitching……..	X	-	-
	Not one continuous length……	-	-	X
	Not stitched to the front side of flag……..	-	-	X
	Not evenly attached around the flag……..	-	-	X
	Stitching of fringe not through center portion of the fringe heading……	-	*	-
	One half of fringe not visible on reverse side of flag.			-
Heading-	Lower end of line not passed through snap hook and sewed to itself……	X	_	-
	Length of zigzag stitching not as specified……	-		X

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	TABLE V DEFECTS CONTINUED			
Type I, Class 1 or 1A or 3, as applicable	Ends of heading not closed by two rows of stitching up to distance line…….	-	*	-
	Free end of distance line not wrapped, fused or taped as specified……	-	*	-
	Sides and ends of duck headings out of alignment…	-	*	-
	Grommet clinched excessively tight, cutting fabric...			
	Grommet improperly set, e.g., Clinched too loosely allowing grommet to rotate……	-	*	-
	Sides and ends of duck out of alignment……	-	*	-
	Grommets clinched excessively tight, cutting fabric.			
	Grommets improperly located……	X	–	-
	Sides and ends of duck heading not properly aligned……	-	*	-
	Grommet improperly set, e.g., Clinched too loosely allowing grommet to rotate……	X	–	-
	Leather tab not positioned and attached as shown on drawing……..	-	*	-
Heading- Type I, Class 2	Sheeting all or partially missing……	-		
	Heading improperly formed, e.g., both edges of the sheeting or hoist end of flag not caught entire length by heading seam…….	-	*	-
Heading- Type I, Class 4	Leather tabs not positioned and attached as shown on drawing……..		*	-
Type II	Heading seam not formed on back of flag…….	-	*	-
	Tie tape missing…….	-		
Heading- Type III, Class 1	Tie tape not positioned or attached as specified……	-		
	Hemmed heading not as shown on drawing……...	-	-	-
Heading- Type III, Class 2	Leather tab not positioned and attached as shown on drawing……..	X	-	X
Marking	Missing, incorrect, illegible, text incomplete, marking in wrong location, size of lettering not as specified or color marking not specified……	-	*	-
	Marking not permanent ink when required…….	-	*	-
Label	Text incomplete or incorrect, not attached as specified, not in location specified or not in position specified…….	-	*	-
	Marking not permanent ink when required……	X	-	-

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4.5.4.1.2 Examination of flag or pennant for defects in dimension.

TABLE VI. DIMENSION DEFECTS

EXAMINE	DEFECT	CLASSIFICATION		
		MAJOR	(*)	MINOR
Overall length (minus heading): Overall width of flag or pennant	Not within the specified tolerance (± 5 % or ± 5/8 inch (1.59cm) per foot……	X	-	-
Width of heading:	Varies from specified finished width by:			
	-- 1/2 inch (1.27cm) or more from specified width....	X	-	-
	--more than 1/4 inch (0.64cm) but less than 1/2 inch (1.27cm) specified width……	-	-	X
Design (emblems, stars, insignia)	Dimension of design or any part thereof out of balance or not in symmetrical alignment (check by comparing approved samples or drawing):			
	-- seriously affecting serviceability or appearance…	X	-	-
	-- affecting serviceability or appearance but not seriously……..	-	-	X

4.5.4.1.3 Inspection levels and acceptable quality levels (AQLs).

The inspection levels and acceptable quality levels expressed in defects per hundred units shall be as follows:

	Inspection Level	AQL Major	AQL Total
For defects applicable to 4.5.4.1.1	I	4.0	10.0
For defects applicable to 4.5.4.1.2	S-3	6.5	15.0

4.5.5 Examination of count of flags and pennants in intermediate containers. Flags and pennants packaged for shipment shall be examined to determine conformance with package marking and specified quantity. The sample unit for this examination shall be one box (interior package). Any box containing less than the specified or marked quantity of flags and pennants shall be classified as a defect. The inspection level shall be s-2 and the AQL shall be 4.0 defects per hundred units.

EXAMINE	DEFECT
Marking, exterior and interior	Omitted, incorrect, illegible or improper size, location Sequence or method of application.
Materials	Any component missing. Any component damaged Affecting serviceability.

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Workmanship	Inadequate application of components, such as Incomplete closure of case liners, container flaps, loose or inadequate strapping. Bulging or distortion of containers.
Weight	Net weight exceeds the requirements.

5 PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6 NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The flags and pennants covered by this specification are intended to be used by individuals and military organizations for identification or signaling.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. The title, number, and date of the applicable specification sheet (See 1.2 and 2.2.1).
c. When first article is required (see 3.2).
d. Ribbon required (See 3.4.8).
e. Type and style service ribbon required (See 3.4.13).
f. Selection of applicable presentation packaging of decorations and medals (See 3.9.1 and 3.9.2).
g. When special marking is required (see 3.11.1).
h. Packaging requirements (See 5.1)

6.3 Naphthalene flake requirements. The naphthalene flake requirement (see 5.1) is not applicable when Government furnished material of wool or part wool content has been moth treated.

6.4 Samples. For access to guide samples, address the contracting activity issuing the Invitation for bids.

6.5 Screen printing. With screen printed applications, experience has shown the type of colorants which provide a degree of colorfastness equal to the basic shades of the flag or pennant are derived from the following classes.

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(a) Cellulosic fabrics; vat dyed in all cases except for U.S. Army scarlet, U.S. Army Maroon and other reds which can be matched only with napthol dyes.

(b) Cellulose reactive dyes and resin-bonded pigment systems may be used successfully in lieu of vat dyes and napthol dyes.

(c) For nylon and nylon-wool buntings, acid and neutral premetallized types have been used.

6.6 Headings. Headings covered by this specification are as shown on the drawing and/or figures below:

Headings	Drawing	Figure
Type I		
Class 1	5-1-45A	Figure 1
Class 1A	5-1-45H	Figure 2
Class 2	5-1-45B	Figure 3
Class 3	5-1-45G	Figure 4
Class 4	5-1-45D	Figure 5
Type II	5-1-45E	Figure 6
Type III		
Class 1	5-1-45F	Figure 7
Class 2	5-1-45F	Figure 7

6.7 First article. When first article inspection is required, the contracting officer should provide specific guidance to offer or whether the item should be preproduction sample, a first production sample or a standard production item from the contractor's current inventory as specified in 4.4. The first article should consist of one completed flag or pennant. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. The invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or tests, must furnish evidence with the bid that prior Government approval is presently appropriated for the pending contract.

6.8 Subject term (key word) listing.

Embroidered
Bonnaz
Screen printed
Applique
Distance line
Heading

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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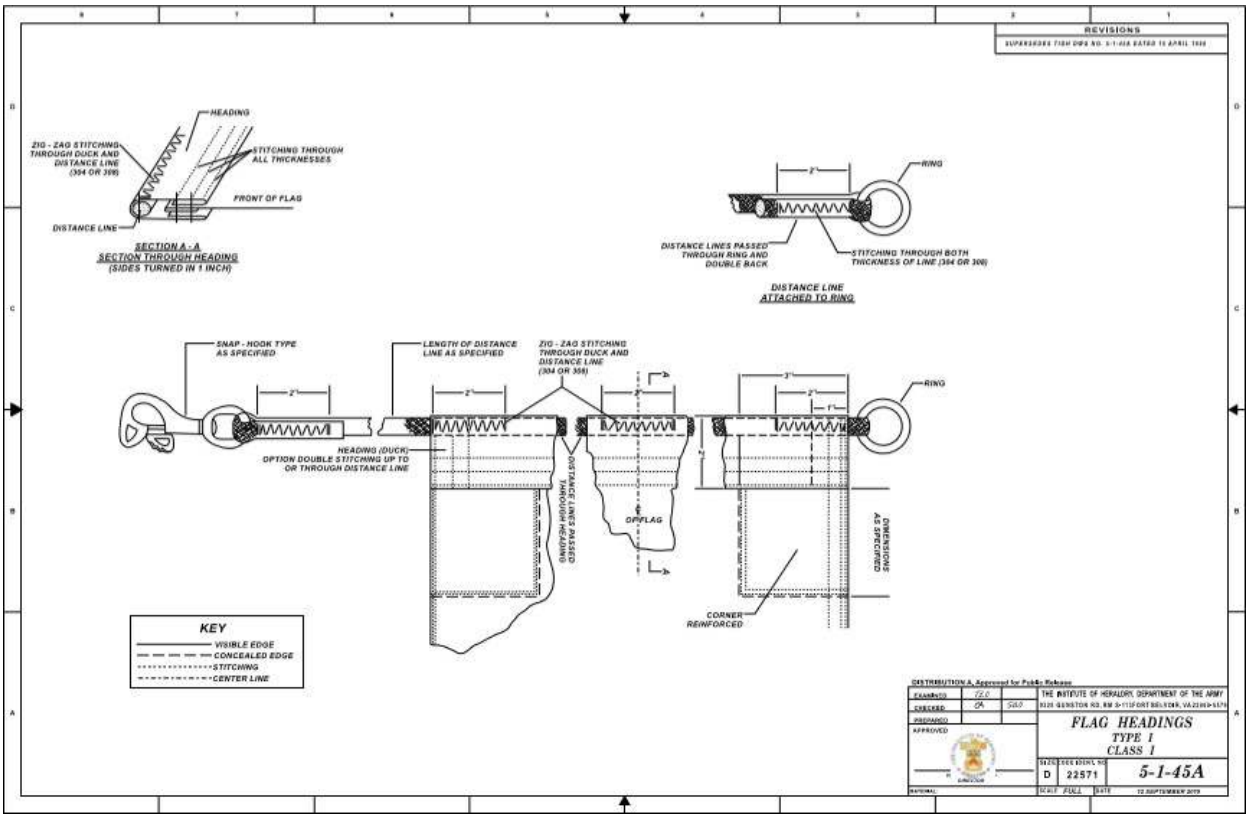


FIGURE 1
TIOH Drawing 5-1-45A Flag Heading Type I Class1
With Distance line (Except for Acrylic Flags)

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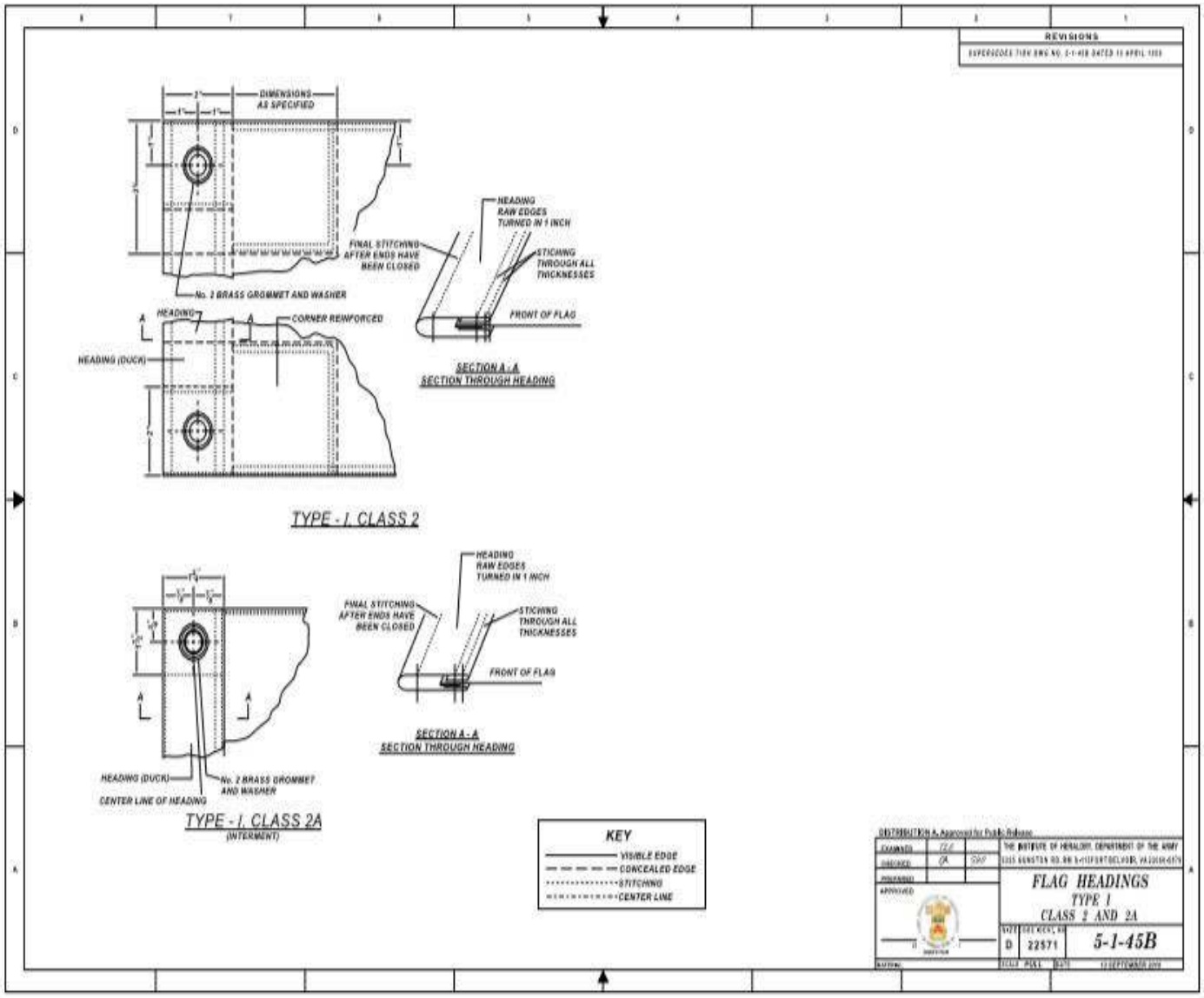


FIGURE 2
TIOH Drawing 5-1-45B Flag Heading Type I Class 2
With Grommets

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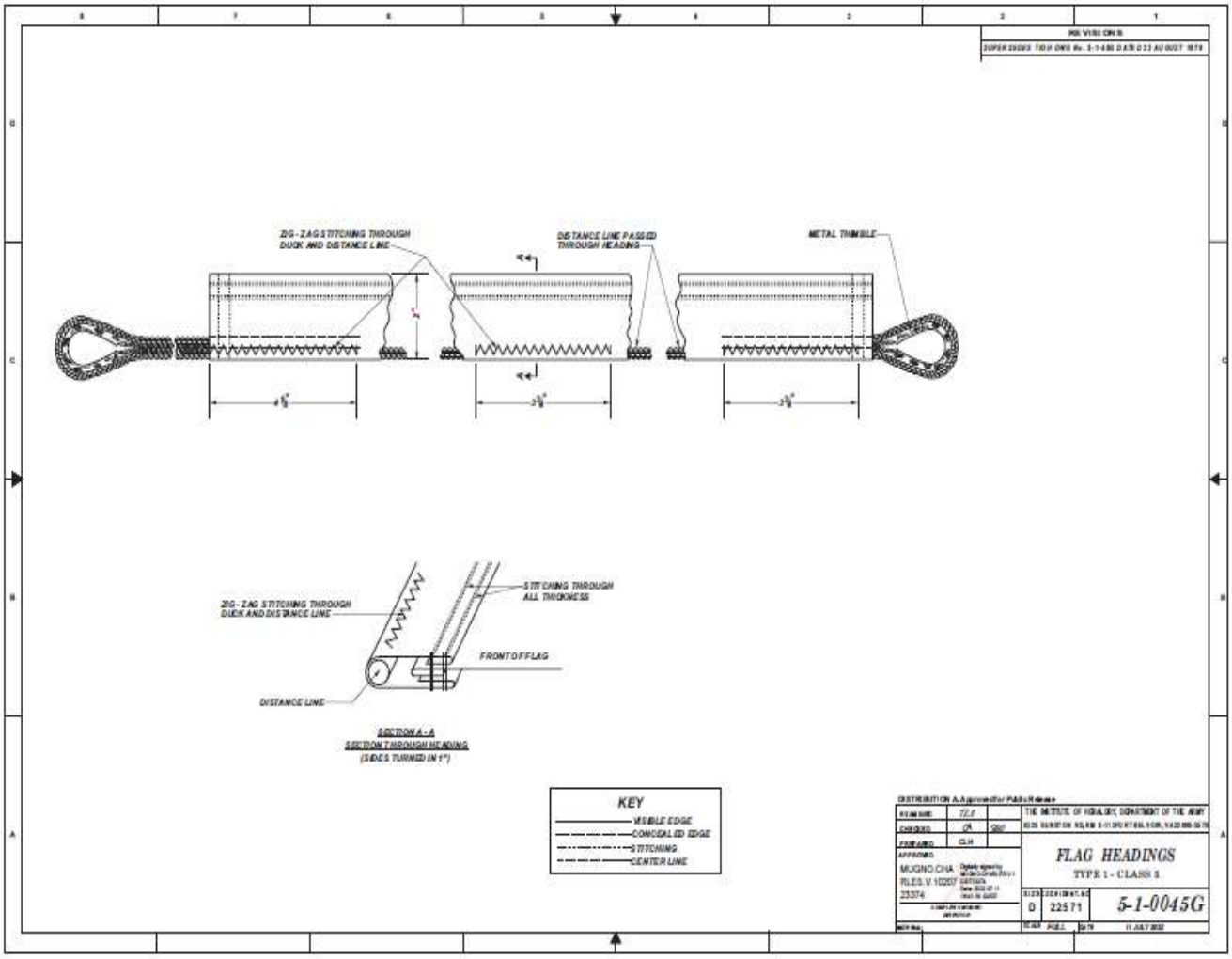


FIGURE 3
TIOH Drawing 5-1-45G Flag Heading Type I Class 3
With Distance Lines and Thimbles

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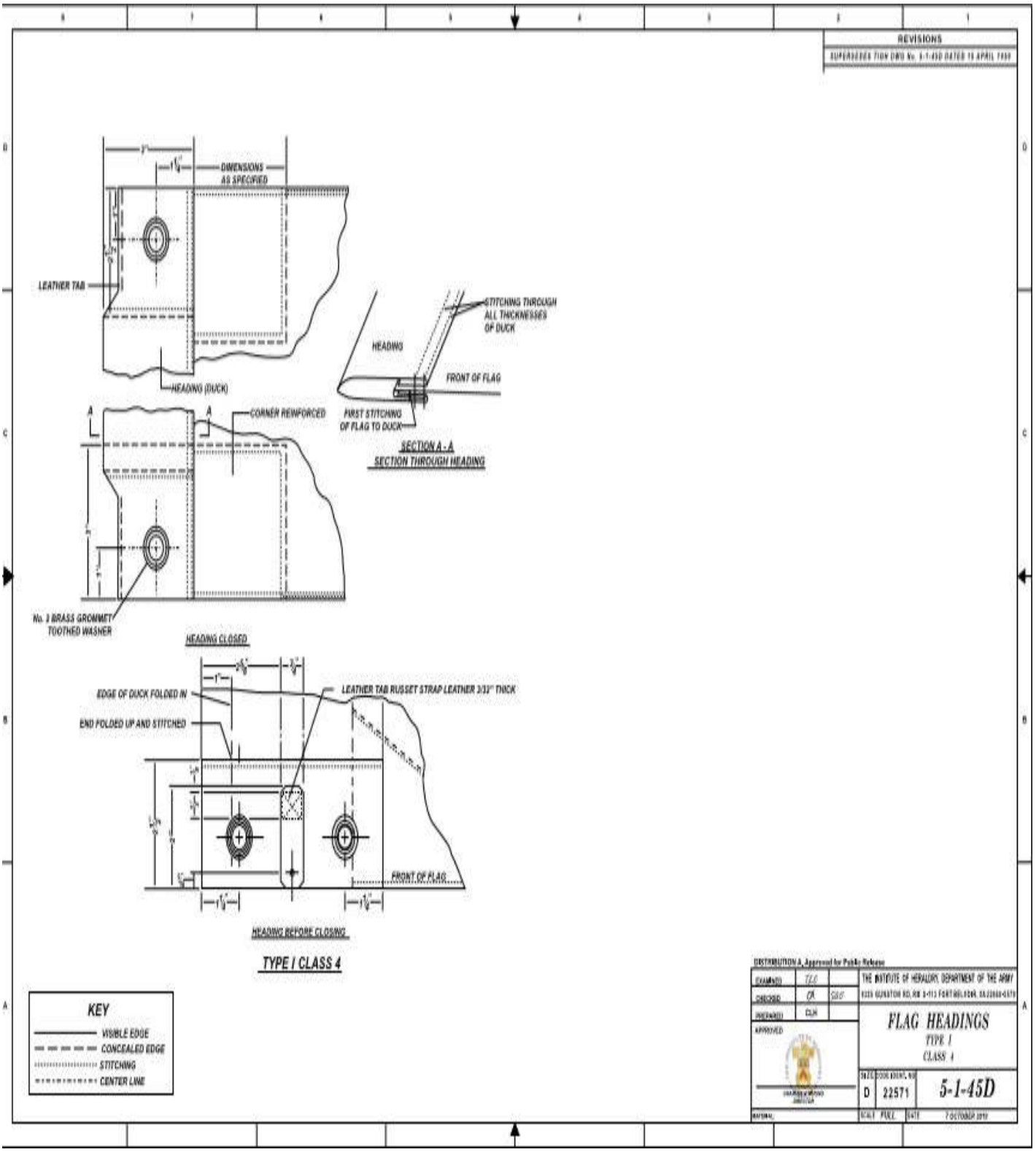


FIGURE 4
TIOH Drawing 5-1-45D Flag Heading Type I Class 4
With Grommets and Leather Tabs

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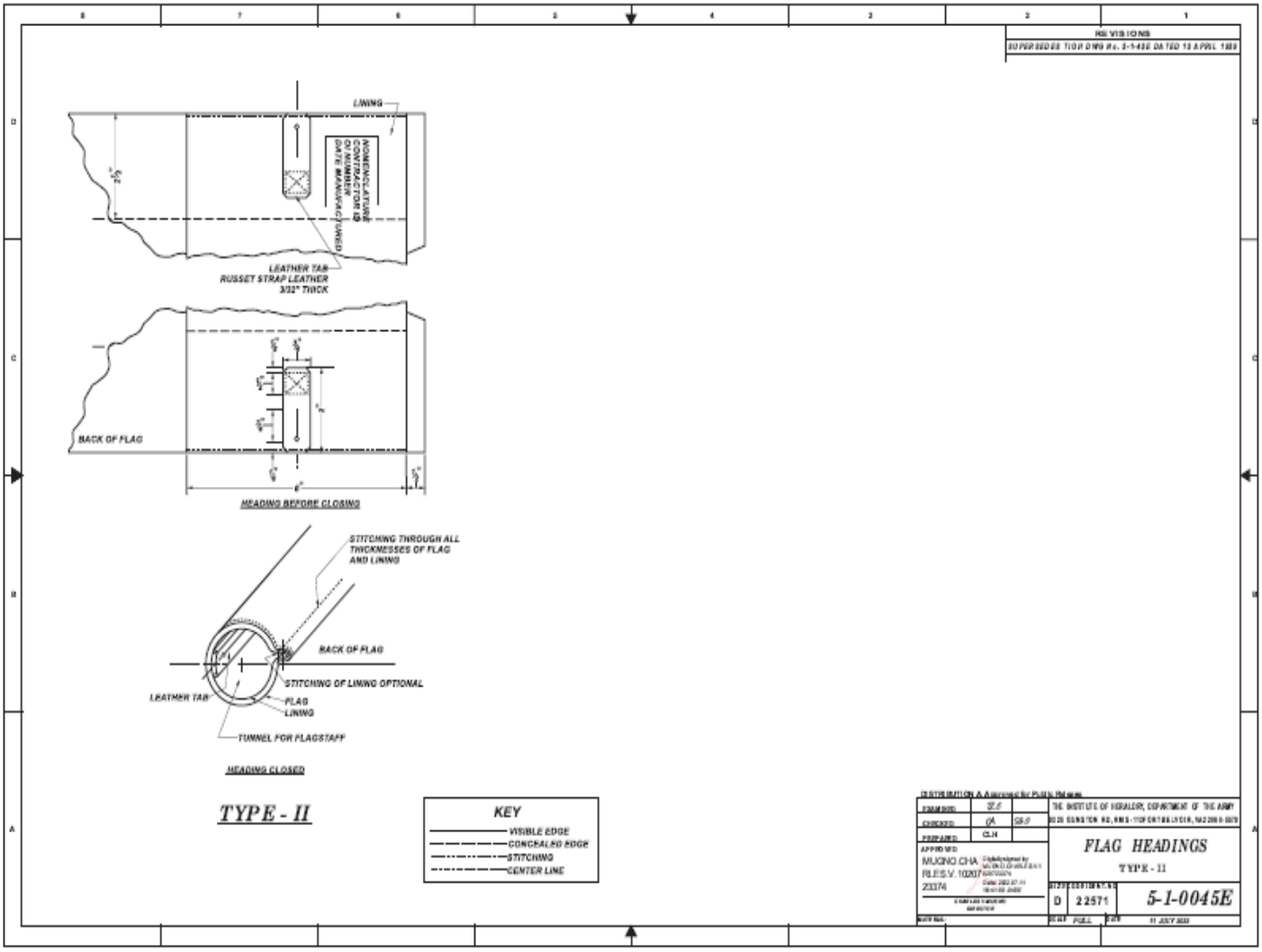


FIGURE 5
TIOH Drawing 5-1-45E Flag Headings Type I
Tube and Tunnel Type (with Cotton Sheeting Lining and Leather Tab)

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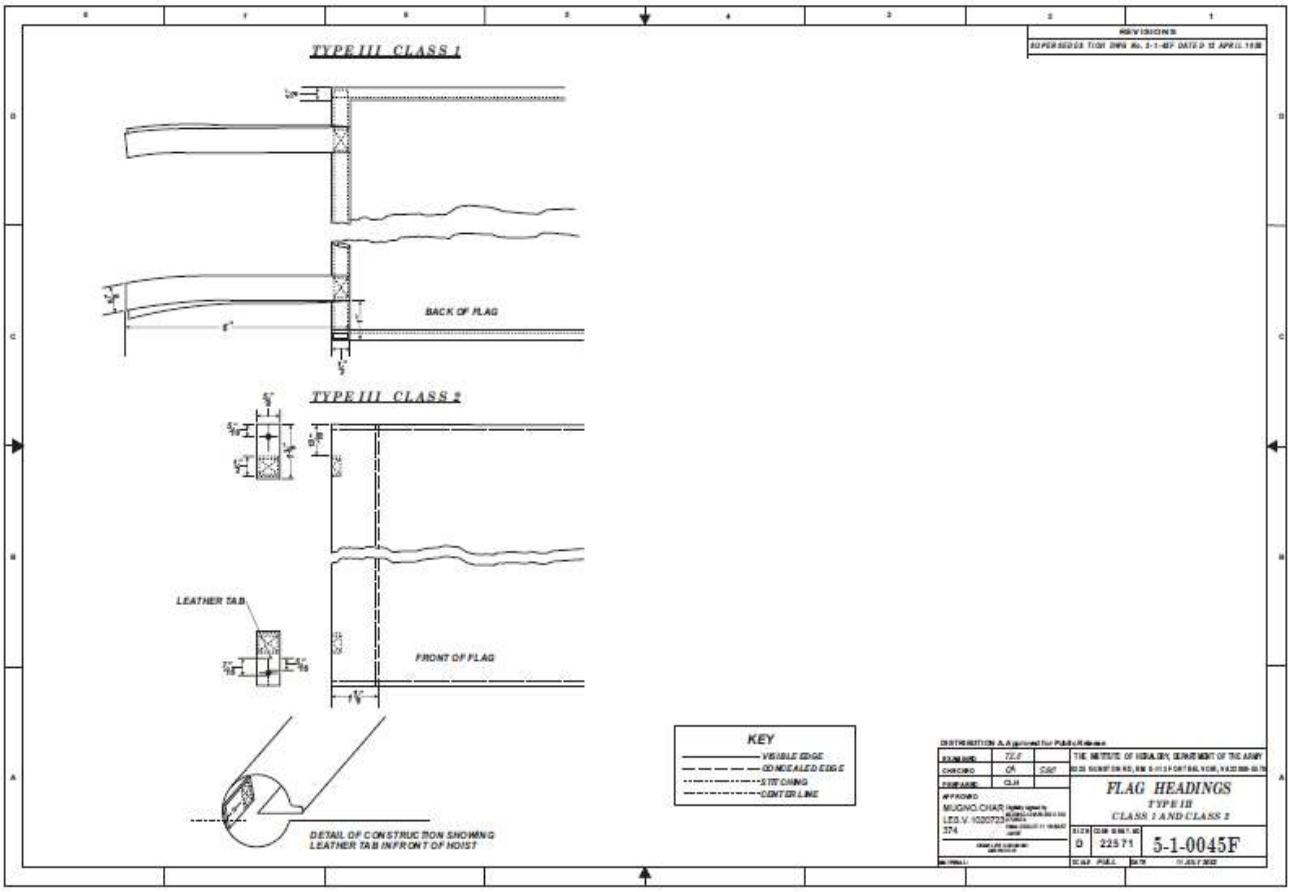
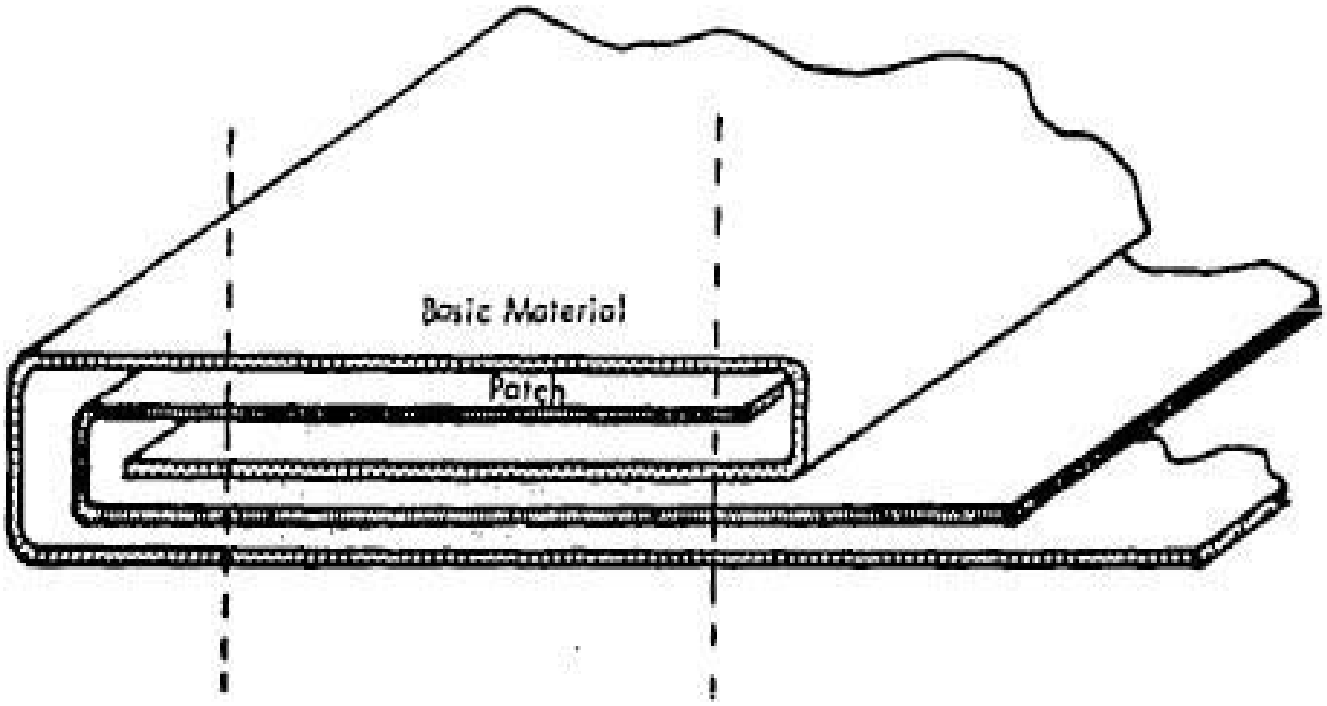


FIGURE 6
TIOH Drawing 5-1-45F Flag Headings Type III Class 1 and Class 2
Type III Class 1 With Tapes

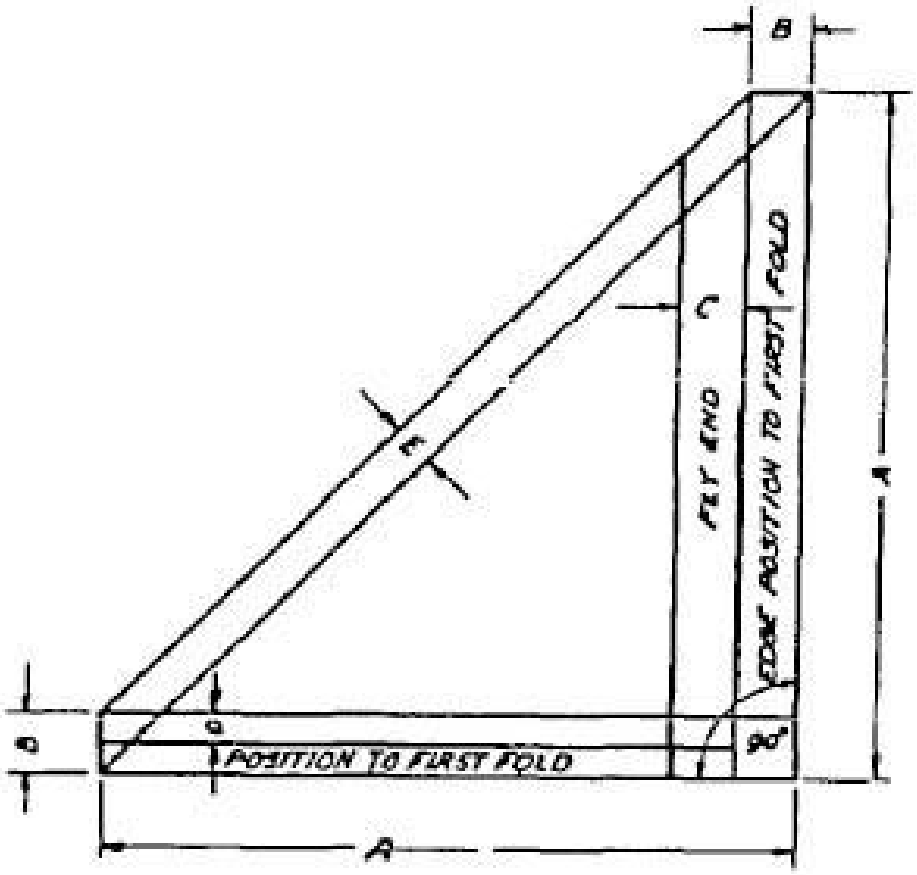
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SEWING OF REINFORCEMENT PATCHES TO THE FLY CORNERS OF FLAG

FIGURE 7

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REINFORCEMENT PATCH

FLAG SIZE	DIMENSIONS IN INCHES				
	A	B	C	D	E
3½	10¼	¾	¾	⅜	½
4	9¼	¾	¾	⅜	½
6	6¼	¾	¾	⅜	½

FIGURE 8

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Custodians:
Army - IH
Air Force - 11
Navy - NU

Preparing activity:
Army - IH

Review activities:
Army - GL
Navy - MC, CG1
DLA - CT

(Project No. 8345-2024-005)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at https://assist.dla.mil.

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